Annotation Reliability in Manglish Sentiment Resources: A Review of Labelling Practices and Evaluation

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Article history

Received:
xx xxx 2023

Received in revised form:
xx xxx 2023

Accepted:
xx xxx 2023

Published online:
xx xxx 2023

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Abstract

Sentiment analysis of Manglish, or Malay-English code-switched text, needs reliable labels for training and evaluation. This review examines the challenges that affect reliable annotation and evaluation of Manglish sentiment data. Peer-reviewed studies were selected from the ACL Anthology, IEEE Xplore and Scopus. The review discusses three language features that make annotation difficult: switching between languages, context-dependent particles such as lah, and non-standard spellings. It then compares manual annotation, sentiment tools and generative models as labelling approaches. Four Malaysian resources are assessed against four criteria: at least 10,000 labelled messages, a class distribution not dominated by one label, social-media language, and direct evidence of label reliability. None meets all four. Two resources report no inter-annotator agreement coefficient. The other two report Cohen's kappa, but only for two-category tasks. MESocSentiment was labelled by majority vote among three sentiment tools, and 78.23% of its labelled messages are neutral. Of six manually examined examples published with the dataset, three contain positive Malay expressions labelled neutral. These examples indicate that further validation is needed. A neutral label may mean that a sentiment tool failed to recognise the sentiment, or that the message is genuinely ambiguous. Majority voting cannot correct an error shared by most of the sentiment tools. Cohen's kappa can also be difficult to interpret when one label dominates. Future studies should document how labels were produced, retain individual labels, evaluate agreement under class imbalance, develop annotation guidance for particles and ambiguous cases, and examine whether annotators and sentiment tools make the same errors.

Keywords: *sentiment analysis, Malay-English code-switched text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is a research field that aims at identifying and analysing the sentiment of unstructured content [1]. Organisations use it to analyse opinions and sentiments in application areas such as consumer products and services [2], public health [3], political discourse [4] and financial services [5]. However, language resources and NLP systems are concentrated on a small number of high-resource languages [6], and many language models are trained mainly on monolingual data [7]. This becomes a problem when a single utterance alternates between two or more languages, a phenomenon known as code-switching [7, 8]. A lexicon-based analyser or a monolingual language model may

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detect sentiment expressed in one language but not in the other, particularly when sentiment-bearing words or linguistic patterns from the second language are not represented or adequately learned. As a result, performance on code-switched text can be lower than performance on the same tasks in a monolingual setting [7].

Both sentiment tools and language models are considered here. In this review, the term sentiment tool refers to systems that directly assign sentiment labels, such as lexicon-based analysers. The term language model is used specifically for pretrained language models, which are trained on large amounts of text and can then be fine-tuned for a downstream task such as sentiment classification [9].

The difficulty of analysing code-switched text is particularly relevant in the Malaysian context. Malay-English code-switching is common in Malaysian social media [10], and a corpus of 709,012 X posts from 50 Malaysian public figures whose posts are mostly Malay-English code-switched has been compiled [11]. In this review, the term Manglish is used specifically to refer to Malay-English code-switched language. Manglish text has recurring switching patterns [12] and informal spellings, such as x for tidak, camne for macam mana [13]. Because such non-standard forms may be poorly represented in standard Malay and English language resources, such as sentiment lexicons and pretrained language models, systems relying on these resources may have difficulty recognising their meaning [14].

Part of the difficulty of processing Malay-English code-switched text stems from how multilingual language models are built and trained. Multilingual pretrained language models such as mBERT [9] and XLM-RoBERTa [15] are trained on corpora that span many languages: Wikipedia in the case of mBERT and Common Crawl in the case of XLM-RoBERTa. These corpora consist largely of monolingual documents, so such models may receive little or no exposure to code-switched text during training [7]. Pretraining on several languages does not necessarily guarantee high-quality representations of text in which those languages appear together [16]. Covering more languages also places demand on model capacity, particularly when the total number of model parameters is fixed [15]. Under this condition, adding more languages initially benefits low-resource languages through positive transfer, but beyond a certain point, performance begins to degrade as model capacity is distributed across more languages, an effect Conneau et al. [15] term the curse of multilinguality. Further training on code-switched examples can improve performance [17], but the results vary by language and training conditions [18].

Model design and pretraining data account for only part of the difficulty in analysing sentiment in Malay-English code-switched text. Another source of difficulty is the quality of the annotated labels in training and evaluation datasets. In supervised sentiment analysis, each training example carries a label that specifies its sentiment class [2]. Incorrect labels can introduce label noise, potentially causing the model to learn misleading patterns from the training data, which reduces predictive accuracy [19]. Labels also play a critical role during evaluation, where they serve as the ground truth against which predictions are scored. Errors in evaluation labels can make reported results misleading and may affect the relative comparison of competing models [20]. Label quality is therefore only one requirement for reliable evaluation. Bowman and Dahl [21] argue that a benchmark

should also be a valid test of the task, offer adequate statistical power, and discourage the use of systems with potentially harmful social biases. Labelling Manglish presents additional challenges. Malaysian social media users tend not to follow standard abbreviation rules, so annotators must interpret non-standard spellings and abbreviations [14]. Annotators must also interpret English words respelled according to Malay pronunciation, such as kompem for "confirmed" and hensem for "handsome", which Kasmuri and Basiron [22] describe as Malayanization words.

These concerns motivate a closer examination of the resources that are currently available. This review examines four Malaysian Malay-English code-switched sentiment datasets: MY-EN-CS, SentiBahasaRojak, MyCovid-Senti, and MESocSentiment [10, 22, 23, 24]. For a resource to support sentiment analysis across a broad range of topics, rather than a single domain such as public health, the following four desirable criteria are adopted by this review: at least 10,000 labelled messages, a class distribution that is not dominated by one label, everyday social-media language, and direct evidence that the labels are reliable. Direct evidence here means a reported inter-annotator agreement coefficient computed on the same label set that the released dataset uses. These are the criteria used in this review, rather than universal requirements for every dataset. When evaluated against these criteria, none of the four resources meets all of them.

The first gap concerns class distribution. MESocSentiment is the largest of these resources in terms of the number of labelled messages, but 78.23% of its labelled messages are classified as neutral [24]. This class distribution warrants further investigation. Of the six manually examined examples published with the dataset, three contain Malay expressions with positive polarity that were assigned a neutral label [24]. These examples need further investigation, although they do not show the overall error rate.

The second gap concerns evidence of label reliability. The four resources provide limited evidence of agreement between annotators. MyCovid-Senti and MESocSentiment do not report an inter-annotator agreement coefficient [10, 24]. MY-EN-CS reports agreement for the two-category distinction between opinionated and factual messages, whereas SentiBahasaRojak reports agreement for positive versus negative sentiment. Neither reports an agreement coefficient for the three sentiment labels of positive, negative and neutral [22, 23].

Inter-annotator agreement can be assessed by having multiple annotators independently label the same messages. An inter-annotator agreement coefficient is then calculated. Cohen's kappa (κ) measures agreement between two annotators while accounting for agreement expected by chance [25]. When more than two annotators are involved, generalised coefficients are available, including Fleiss's multi-π, multi-κ and Krippendorff's α [26]. However, Cohen's kappa can be sensitive to the marginal distribution of labels, making the coefficient harder to interpret when one class is much larger than the others [27, 28, 29]. A dataset with many neutral labels therefore warrants careful examination of both label quality and inter-annotator agreement. The issues described so far concern the quality of existing labels. Resource availability is another limitation, as publicly accessible datasets of Manglish social-media text are scarce [11]. When only a few resources

are available, the quality of the labels in those resources becomes particularly important.

These gaps frame the research question addressed in this review: what challenges affect the establishment of reliable annotation and evaluation of Manglish sentiment data? The review addresses this question through three aspects: the linguistic features that complicate annotation, the approaches used to assign labels, and the evidence reported for label quality. The purpose is to assess the evidence supporting existing labels and identify the reliability checks that should be reported in future studies. Existing reviews have examined sentiment classifiers and sentiment analysis more broadly [1, 30], whereas this review focuses on the annotation and evaluation of Manglish sentiment data rather than proposing a new system.

To address this question, the review makes three contributions:

- It compares manual annotation, existing sentiment tools and generative models as approaches to sentiment labelling, focusing on the effort they require for labelling and the evidence they provide for label reliability.
- It explains the difference between missed sentiment cues and differences arising from reasonable alternative interpretations, and examines the reliability evidence for four Malay-English code-switched sentiment resources.
- It examines the relationships among linguistic challenges, labelling methods and class imbalance, and identifies priorities for future research.

The review is organised as follows. Section 2 describes the literature selection. Section 3 discusses linguistic challenges, followed by annotation approaches in Section 4 and label reliability evaluation in Section 5. Section 6 synthesises these issues and outlines research priorities. Section 7 concludes the review.

2. Review Methodology

The literature was identified through structured searches of the ACL Anthology, IEEE Xplore and Scopus. The search strategy combined terms from three groups. Language-related terms included code-mixed, code-switching, Manglish, Bahasa Rojak and Malay-English. Task terms included sentiment analysis, sentiment polarity and opinion mining. The final group focused on datasets and label quality, using terms such as corpus, annotation, inter-annotator agreement and label quality.

The main search covered studies published from 2019 to 2026. Earlier studies were considered when they were cited in the papers found during the search. Studies on agreement measures were not limited to this period because several commonly used measures were introduced before 2019, such as Cohen's kappa [25].

Four criteria were used to determine inclusion. Studies were included if they: (1) examine Malaysian Malay-English text, or propose or evaluate methods for assigning labels, measuring inter-annotator agreement, or identifying annotation errors; (2) report the annotation procedures and any methods used to assess label quality; (3) were published in a peer-reviewed journal or conference; (4) address annotation procedures, label quality, or inter-annotator agreement. Preprints without a peer-reviewed version were excluded.

Studies that only reported classifier accuracy on an existing dataset were excluded. Such studies used the supplied labels to evaluate model performance, but do not necessarily assess the reliability of those labels. This review focuses on the evidence supporting the reliability of the labels themselves.

3. Linguistic Challenges for Annotation

The selected studies highlight three features of Manglish that make sentiment annotation difficult: switching between languages, context-dependent discourse particles, and non-standard spelling.

3.1. Sentiment across Language Boundaries

In Malay-English code-switched text, both languages can appear within a single sentence [12]. In their study of Malaysian classroom speech, Treffers-Daller et al. [12] found frequent switching within clauses. However, the study does not establish how common such within-clause switching is in social-media writing.

This presents a challenge for sentiment annotation because sentiment-bearing expressions may occur in either language. This can create difficulties for sentiment tools when sentiment expressed in one language is not adequately captured. Bilingual annotators may be able to interpret both languages, but linguistic comprehension alone does not guarantee consistent annotation. However, bilingual annotators still need guidance when a message contains multiple targets associated with different sentiments, because ambiguous annotation guidelines may lead annotators to assign inconsistent labels [31].

3.2. Particles and Context

Particles such as lah can affect the pragmatic tone of an utterance without changing its basic content [32]. Tay et al., as cited in Treffers-Daller et al. [12], suggest that lah can change an utterance from a highly assertive tone into a polite request or a friendly encouragement. AlAfnan et al. [32] use "Don't disturb la, he's busy" as an example of a softened instruction. The spelling la in this example also illustrates the variation discussed in Section 3.3. These studies examine classroom speech and television dialogue, so they provide evidence about the functions of particles rather than how often each use appears on social media.

The particle lah therefore cannot be given a fixed positive or negative meaning. Depending on the context, the same particle may contribute to different pragmatic interpretations, such as politeness or irritation. Annotation guidelines should explain how to consider this pragmatic context when assigning a positive, negative, or neutral sentiment label.

3.3. Spelling Variations

Users often write Malay words in shortened or non-standard forms [13]. Maskat and Rahman [13] report forms such as jgn for jangan and mcmana for macam mana. These forms may be poorly represented in sentiment lexicons and pretrained language models, so systems relying on these resources may have difficulty

recognising their meaning. One possible approach is spelling normalisation, which maps an informal form back to its standard spelling. In Maskat and Rahman's [13] experiment, the highest precision was 0.43. This means that 43% of the proposed matches were correct. However, the evaluation excluded English words and code-mixed posts. The result therefore does not directly evaluate performance on Malay-English code-mixed text.

Negation makes spelling variation particularly relevant to sentiment. As discussed in Section 1, x can represent tidak. If a tool or model fails to recognise this negative form, it may misinterpret the sentiment conveyed by the utterance. A spelling-normalisation method should therefore be evaluated not only on whether it produces the standard spelling but also on whether it preserves the original sentiment.

4. Annotation Approaches and Their Limitations

4.1. Human and Automatic Labelling

A human annotator with proficiency in Malay and English may be better positioned to interpret locally used language. A human annotator can consider the surrounding context, informal spellings and particles, when interpreting a message as a whole. However, manual work is time-consuming and requires appropriate language proficiency [24]. For example, Kasmuri and Basiron [22] used human annotators to label blog sentences as opinionated or factual when building MY-EN-CS, a subjectivity corpus. Kong et al. [10] used three independent annotators and written guidelines to annotate 11,568 tweets. Human judgement does not by itself establish label reliability; where multiple annotators are involved, inter-annotator agreement provides direct evidence of annotation consistency.

Dataset construction may also involve reusing or automatically transforming existing text rather than collecting naturally occurring code-mixed data. In SentiBahasaRojak, the product and movie subsets were created by automatically translating parts of existing Malay sentences into English or Chinese, so the resulting code-mixed text was automatically generated rather than naturally occurring user-generated text [23]. The stock-market posts were collected from Malaysian financial and stock-market websites and annotated by five experts as positive or negative [23]. The reported inter-annotator agreement was calculated only for the stock-market subset. It should therefore be interpreted as evidence for that subset, rather than for the entire dataset.

Sentiment tools can assign labels more quickly than manual annotation [24]. Mohd Shamsuddin et al. [24] used TextBlob, VADER and pysentimiento to annotate MESocSentiment. The label assigned by the majority of the tools was used as the final label. The authors describe the overall dataset construction as semi-automatic, although the sentiment labels themselves were assigned automatically. The reliability of these automatically assigned labels on Malay-English code-switched text therefore requires independent verification.

Generative models offer another way to label text, but their performance needs to be evaluated for the specific language and task. Jadhav et al. [33] found that GPT-

4o and Llama 3.1 405B performed worse than a fine-tuned BERT model on four Marathi classification tasks, including sentiment analysis. This finding does not establish how the models would perform on Malay-English code-switched text in Malaysia, and none of the studies reviewed here tests generative models as annotators for Malay-English sentiment.

4.2. Interpreting Neutral Labels

Automatic labelling may produce a large neutral class, because a sentiment tool that does not recognise a sentiment-bearing expression may assign a neutral label. MESocSentiment, which was labelled automatically, contains 19,714 posts, of which 78.23% are labelled neutral, 16.33% positive and 5.44% negative [24]. A large neutral class does not by itself indicate that the labels are incorrect. Possible explanations include the topics represented in the dataset, genuinely ambiguous sentiment, and limitations in the sentiment tools used for automatic labelling. The key question is whether some posts were labelled as neutral because the tools failed to recognise a sentiment.

One way to assess whether the neutral distribution warrants closer examination is to compare it with other Malaysian sentiment datasets. The 78.23% neutral proportion in MESocSentiment is higher than that reported for the Malaysian COVID-19 Twitter dataset of Kong et al. [10], in which three annotators labelled 11,568 tweets, of which 23.6% were labelled neutral. The two datasets cover different content, so their class distributions are not directly comparable. The difference nevertheless suggests that the neutral proportion in MESocSentiment warrants further investigation.

One possible cause of incorrect neutral labels is limited language coverage. VADER, one of the three tools used, computes a compound sentiment score from lexical and rule-based sentiment cues and assigns neutral when the resulting score falls within its neutral range [34]. If a Malay expression is not represented in its sentiment lexicon, that expression may not contribute to the score. A message containing sentiment-bearing expressions may then receive a neutral label.

The examples published with MESocSentiment provide some evidence consistent with this explanation. Of the six manually checked examples presented by Mohd Shamsuddin et al. [24], three contain Malay expressions that appear to convey positive sentiment but were labelled neutral. The original authors identified these three tweets as incorrectly labelled when they checked them manually. The first describes a village in Penang as memang cantik sangat, meaning really very beautiful. The second, Selamat deepavali buat followers bikers ranger, wishes followers a happy Deepavali. The third, Syabda raih gelar juara international series, reports that an athlete won a championship title. These examples are consistent with the possibility that the tools may fail to capture sentiment expressed in Malay. However, six examples are not enough to estimate the overall labelling error rate of the dataset. The high proportion of neutral labels therefore cannot be attributed solely to missed sentiment cues. Other factors may also contribute to neutral labels.

Not every neutral label reflects a missed cue. Plank [35] argues that variation between annotators is not always noise, since a message may have more than one

plausible reading. In a study of natural language inference, disagreement between raters persisted even when 50 judgements were collected for each item [36]. Providing more context did not reduce this disagreement, and in their experiment, disagreement increased as raters were shown more context. These studies do not examine Malaysian data, but they highlight an annotation issue that is not explicitly addressed in MESocSentiment. Annotation guidelines should distinguish genuinely neutral messages from messages for which the sentiment is ambiguous or difficult to determine.

4.3. Limits of Majority Voting

The use of multiple labelling tools and majority voting does not necessarily resolve this ambiguity. Majority voting gives each voter equal weight and cannot correct a label when most voters make the same error [37]. In MESocSentiment, the three voters are sentiment tools rather than human annotators, so majority voting cannot correct an error shared by most of the tools. Each of the three tools labels a majority of the posts as neutral, as shown in Table 1.

Table 1. Neutral Labels in the 19,714 Retained MESocSentiment Posts, Calculated from Mohd Shamsuddin et al. [24]

Tool	Neutral posts	Proportion
TextBlob	14,305	72.56%
VADER	13,115	66.53%
pysentimiento	16,916	85.81%
Majority vote	15,422	78.23%

The proportion of neutral labels differs across the three tools. pysentimiento produces the highest proportion of neutral labels. It supports English, Spanish, Portuguese and Italian, and was selected by Mohd Shamsuddin et al. [24] because it is trained on English tweets. Malay is not among the languages for which it was trained. The neutral-label proportion alone does not establish whether pysentimiento produces less reliable labels. The label counts in Table 1 also do not reveal whether the tools make errors on the same posts. If at least two out of three tools assign the same incorrect label to a post, the resulting majority label will also be incorrect. Independent review by bilingual annotators would therefore be needed to determine whether the tools make errors on the same posts.

The authors removed 512 of the original 20,226 posts when all three tools assigned different labels [24]. Each removed post therefore had one neutral vote, one positive vote and one negative vote. The number of neutral posts stayed at 15,422 while the total number of posts decreased, causing the neutral proportion to increase from 76.25% to 78.23%. Removing posts with conflicting tool assignments may also exclude difficult examples, although the published counts do not show what these posts contain. Keeping the posts and each tool's labels would allow researchers to examine the disagreements directly [38].

5. Evaluating Annotation Reliability

Examining such disagreements requires measuring agreement, but agreement alone does not establish that the assigned labels are correct. Two annotators may

agree simply because they made the same mistake [26]. Cohen's kappa [25] does not identify errors that are shared by both annotators. It adjusts the observed agreement for the agreement expected by chance. It compares observed agreement with the agreement expected by chance, where chance is estimated from how often each annotator uses each label. Artstein and Poesio [26] discuss a reliability measure proposed by Krippendorff [39], in which values above 0.8 are considered reliable, and values between 0.67 and 0.8 as supporting only tentative conclusions. These ranges should be read as guidance rather than fixed rules, because the task, annotator characteristics, and label distribution can all affect the interpretation of a given coefficient.

Among these factors, label distribution is particularly important when interpreting kappa. When one label is far more common than the others, the agreement expected by chance rises, causing kappa to decrease even when the observed agreement remains high. Di Eugenio and Glass [29] illustrate this with two binary examples with 90% observed agreement. Kappa is 0.80 for the balanced example, but -0.048 for the skewed example. The same issue can arise in sentiment corpora when one sentiment class substantially outnumbers the others. In MESocSentiment, 78.23% of posts are labelled neutral [24]. Observed agreement and kappa should therefore be reported and interpreted together.

Overall agreement should therefore be reported alongside per-class agreement, so that weak agreement on minority classes is not obscured by the dominant class. The individual annotation should also be stored per annotator, not only as label totals, because Cohen's kappa requires the correspondence between each annotator's labels for the same items [29]. No single coefficient gives a complete account of reliability.

Table 2. Reliability Evidence Reported for the Reviewed Resources

Resource	Annotation evidence	Scope and limitation
MY-EN-CS Kasmuri and Basiron [22]	Two bilingual undergraduates; written guideline; reported kappa = 0.83.	Opinionated versus factual judgement. Does not establish three-class polarity agreement.
SentiBahasaRojak Romadhona et al. [23]	Five experts labelled stock posts; reported kappa = 0.77.	Positive versus negative stock labels; coefficient variant unspecified. Other subsets were generated automatically.
MyCovid-Senti Kong et al. [10]	Three independent annotators; written guideline.	Three-class sentiment labels; no agreement coefficient reported.
MESocSentiment Mohd Shamsuddin et al. [24]	Majority vote among three automatic tools.	No human inter-annotator agreement coefficient reported.

Table 2 summarises the evidence reported by each study and the aspects of three-class label reliability that remain unassessed. The studies introducing MyCovid-Senti and MESocSentiment do not report an inter-annotator agreement coefficient. The study of MY-EN-CS reports a kappa value of 0.83 and the study of SentiBahasaRojak reports a kappa value of 0.77, but both coefficients were calculated for two-category tasks, and therefore do not provide evidence of agreement on the three-class positive, negative, and neutral sentiment labels. SentiBahasaRojak also used five annotators, but the study does not state how its kappa value was calculated [23]. Cohen's kappa in its original form measures agreement between two annotators [25]. When more than two annotators are involved, agreement can be averaged over pairs of annotators or measured with a generalised coefficient [26]. Studies involving more than two annotators should therefore report how agreement was calculated and specify the coefficient or variant used. None of this shows that the four resources are unreliable. It shows that the reviewed studies do not provide evidence that label reliability has been assessed for the three-class sentiment distinction.

Table 3 checks the four resources against the four criteria adopted in Section 1. A class distribution is treated as dominated when one label covers more than half of the labelled messages. MyCovid-Senti meets three criteria, but its study reports no inter-annotator agreement coefficient. MESocSentiment is the largest resource, but its labels are dominated by the neutral class and were assigned by sentiment tools, so no inter-annotator agreement coefficient is available. MY-EN-CS and SentiBahasaRojak each contain fewer than 2,400 labelled messages, and their agreement evidence covers only a two-category task or a single subset. None of the four resources therefore meets all the criteria adopted by this review for broad-topic sentiment analysis.

Table 3. The Four Reviewed Resources Against the Four Criteria, Compiled from Kasmuri and Basiron [22], Romadhona et al. [23], Kong et al. [10] and Mohd Shamsuddin et al. [24]

Resource	At least 10,000 messages	Not dominated by one label	Social-media language	Direct reliability evidence
MY-EN-CS Kasmuri and Basiron [22]	No. 2,316	Yes. 52% opinionated, 48% factual	Partly. Blog posts	Yes. for opinionated versus factual only
SentiBahasaRojak Romadhona et al. [23]	No. 2,285	Not reported	Partly. 693 posts written by users	Partly. Stock subset only
MyCovid-Senti Kong et al. [10]	Yes. 11,568	Yes. Largest label 48.9%	Yes	No
MESocSentiment Mohd Shamsuddin et al. [24]	Yes. 19,714	No. 78.23% neutral	Yes	No

6. Research Priorities

As Table 2 shows, the reviewed studies generally provide more detail on label production than on the assessment of label reliability. The following discussion explains how the linguistic, labelling, and evaluation challenges are connected and sets out four priorities for improving annotation and evaluation.

6.1. Interaction Among the Challenges

The challenges discussed in the previous sections are interrelated rather than independent. Manglish text is usually written with non-standard spellings, and its discourse particles, such as lah and meh, signal the speaker's attitude and are highly context-dependent. Sentiment tools trained on other languages may therefore fail to recognise some of these non-standard or context-dependent expressions [24]. Bilingual annotators can interpret such expressions, but recruiting and compensating appropriately qualified annotators can be costly. Mohd Shamsuddin et al. [24] identify annotation cost as one reason for using automatic labelling. However, automatic labelling introduces its own limitations: for example, VADER relies on an English sentiment lexicon [34]. When several sentiment tools fail to recognise the same sentiment-bearing expression, majority voting does not correct the error and may instead propagate it. Posts expressing positive or negative sentiment may therefore be labelled as neutral, which can obscure sentiment-bearing information and further increase the dominance of the neutral class discussed in Section 5.

A highly skewed label distribution has two further implications. The first concerns measurement. When one label dominates, annotators may agree frequently even when some of the agreement is attributable to chance. A low kappa value can therefore be difficult to interpret. It may signal genuinely low inter-annotator

agreement. It may also be influenced by an uneven label distribution [27, 28]. The second implication concerns modelling. Systematic label errors can cause a model to learn misleading patterns from the training data, which can reduce predictive accuracy [19]. Such a model may then be used to label further data, creating the possibility of propagating the same errors, as illustrated by the sentiment analysers used to label MESocSentiment [24]. Without independent verification, such errors may be propagated to a larger dataset without revealing the extent to which its labels are reliable.

Addressing these risks requires quality checks at more than one stage. The priorities presented in the following sections address how labels are produced, validated and retained for subsequent analysis.

6.2. Report How Labels Were Produced

The first priority is to document how the labels were produced so that the dataset can be interpreted and reused appropriately. Dataset releases should identify how the labels were assigned, including whether they were produced by human annotators, automatic tools, or a combination of both, and should provide the annotation guidelines. They should also specify the annotation task and the judgement that annotators were instructed to make. A kappa value calculated for a two-class annotation task does not establish agreement for a different three-class label scheme, yet a single reported value may be presented alongside the released dataset without clearly identifying the task for which it was calculated.

Dataset documentation should also describe the annotators and the context in which the labels were produced. Data Statements offer an established format, covering language variety, data contributors, and the annotators' backgrounds [40]. Such documentation allows researchers to assess the intended scope of a dataset and the contexts in which its use may be limited.

Beyond documenting the annotation context, dataset releases should also report the distribution of labels and preserve the individual annotation decisions. Class distribution should be reported alongside agreement for each class, so that weak agreement on minority classes is not obscured by a single overall coefficient. Retaining each annotator's labels, along with any items excluded because of annotation disagreement, allows later studies to examine the items on which annotators disagreed rather than relying only on the majority label [41]. The 512 posts removed from MESocSentiment illustrate the information that becomes unavailable when conflicting annotation decisions are not retained.

6.3. Evaluate Agreement under Class Imbalance

The class distribution is another factor that should be considered when evaluating annotation agreement. Future studies should examine how agreement measures change as the proportions of positive, negative and neutral labels change. Byrt et al. [28] note that reporting kappa values alone can be misleading when agreement studies are compared, and recommend that researchers also report indicators of bias and prevalence. Di Eugenio and Glass [29] discuss this recommendation for discourse and dialogue annotation and suggest reporting kappa together with a bias-

adjusted value and a prevalence-adjusted value. These approaches were developed for two raters and two categories [28]. None of the four Malaysian resource studies reviewed here reports them, so their use with three-class sentiment labels has not been examined in these studies [10, 22, 23, 24]. Applying these analyses to three-class Manglish sentiment data could help determine how class distribution affects the interpretation of observed agreement and kappa.

6.4. Develop Annotation Guidance for Particles and Ambiguity

Statistical checks address how agreement is measured, but they do not resolve the linguistic ambiguities that cause annotators to disagree. Guidelines provide a direct means of addressing the linguistic challenges identified in Section 3. They should include worked examples showing how the pragmatic function of the same particle varies with context, since lah and kan convey pragmatic or discourse-level cues that no lexicon entry can capture. Mohammad [31] provides a useful framework that distinguishes the opinion holder, the target of the opinion, and the expressed attitude, and Kong et al. [10] adapted this structure for Malaysian tweets. Manglish annotation guidelines could extend this framework with examples of locally used discourse particles and context-dependent interpretations.

Guidelines should also specify how annotators should handle items that remain ambiguous. Requiring annotators to choose between positive, negative and neutral can reduce an item with multiple plausible interpretations to a single label, which may then be treated as annotator error [35]. Allowing annotators to identify such cases separates disagreement caused by vague instructions, which may be reduced through improved guidelines, from disagreement arising from genuine ambiguity in the text.

6.5. Examine Potentially Shared Annotation Errors

Section 5 noted that annotators may agree because they make the same error, and Section 6.1 described how shared limitations in language resources or interpretation can produce consistent but potentially incorrect labels. Neither agreement coefficients nor majority voting can establish whether an agreed label is correct [26, 37], so an independent validation procedure is required to identify potential shared errors.

One possible validation design would present the same sample of Manglish messages to bilingual annotators, to existing sentiment tools, and to generative language models used for sentiment labelling. A separate group of bilingual reviewers would then assign independent human reference labels and provide brief explanations for the cases in which the automated systems disagree or in which their consensus label differs from the independent human label. Weber and Plank [42] found that adding human corrections to automatic error detection improved the identification of annotation errors compared with automatic detection alone. Weber-Genzel et al. [43] similarly used explanations and subsequent validity judgements to distinguish annotation errors from reasonable variation in English textual inference. A similar validation design could be evaluated for Manglish sentiment annotation.

The comparison would show whether majority voting corrects individual errors or preserves the ones shared by the annotators. This is particularly important when humans and automatic tools fail to recognise a sentiment-bearing linguistic feature. In such cases, majority voting cannot resolve the shared error because the same label is supported by the majority of annotators.

7. Conclusion

This review examined the challenges involved in establishing reliable annotation and evaluation of Manglish sentiment data. Three connected factors were identified: (1) linguistic features such as code-switching, particles and non-standard spellings complicate annotation; (2) the approaches used to assign labels may fail to adequately capture these features, especially when automatic tools are used; and (3) the evidence reported for label quality is limited. The four resources reviewed here provide useful Malay-English data, but the evidence supporting their label quality remains incomplete. Some of the studies report no inter-annotator agreement coefficient, while others report one for a different task or for only part of the dataset. Examples reported in the MESocSentiment study include Malay expressions that appear to convey positive sentiment but were assigned neutral labels by the automatic tools. This does not establish an overall error rate, but it indicates that further validation is needed.

Four priorities would strengthen the evidence. Future work should document how labels were produced and retain each annotator's labels, evaluate agreement under class imbalance, develop guidance for particles and ambiguous cases, and examine whether annotators and tools share potential annotation errors. Without these checks, annotation errors may remain undetected and can propagate into models trained and evaluated using these datasets.

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